PROJECT VARUNA (AERO-SENSE)

COMPREHENSIVE TECHNICAL APPROACH & FULL-STACK SYSTEM SPECIFICATION

Smart India Hackathon 2026 | Problem Statement: SIH26078 | Sponsoring Body: MoES / NCMRWF | Team: AETHON

Problem Statement ID:	SIH26078 (Multi-Hazard Warning)	**Team Name:**	Team AETHON
Project Codename:	VARUNA (AERO-SENSE)	**Sponsoring Body:**	Ministry of Earth Sciences (MoES) / NCMRWF
Production URL:	https://varuna-access.vercel.app	**PDF Dossier URL:**	https://varuna-access.vercel.app/VARUNA_SIH26078_Co
Demo Credentials:	ID: OP-8842-MET | Password: IMD@2026#Sec	**Security Posture:**	Zero Client-Side API Key Exposure

SECTION 1: OPERATIONAL PROBLEM DEFINITION & NATIONAL GAPS

Severe weather anomalies in India-ranging from Category-5 super cyclones in the Bay of Bengal to extreme cloudbursts in the Western Ghats and acute heatwaves in the Indo-Gangetic Plains-present immense socioeconomic disruption. The central limitation in current disaster response is not numerical prediction accuracy, but the dissemination interface:

- The 25-50 km District-Scale Warning Fallacy: Conventional alerts blanket entire revenue districts (2,000 to 8,000 sq km). Over 85% of citizens within the alerted zone experience no critical hazard, leading to acute warning fatigue and complacency.
- Massive Avoidable Economic Losses: Untargeted shutdowns of ports, schools, rail lines, and industries cause unnecessary daily losses amounting to tens of crores of rupees in safe sub-districts.
- Warning-Evacuation Disconnection: Standard meteorological advisories recommend evacuation without integrating real road networks, actual shelter capacities, or real-time bottleneck status, forcing local administrators to improvise under emergency conditions.

SECTION 2: COMPLETE TECHNOLOGY STACK & PROGRAMMING LANGUAGES INVENTORY

Project VARUNA is architected on a high-performance, modular cyber-physical stack engineered for zero downtime, sub-second interactive map rendering, and verifiable mathematical precision:

Layer / Domain	Language / Technology	Specific Architectural Role in VARUNA	Version / Protocol
Frontend UI / UX	**JavaScript (ES2023+) / React**	Virtual DOM component lifecycle, custom hooks, real-time telemetry state	React v18.3.1
Build & Bundler	**Vite 5.4 / Node.js 24**	ESBuild native compiling, rapid HMR, optimized chunking & tree-shaking	Vite 5.4.10 / Node 24
Interactive Canvas	**HTML5 Canvas 2D API**	60 FPS hardware-accelerated slippy map tile rendering & flow particles	W3C Canvas 2D
Geospatial & Vector	**D3-Geo & TopoJSON**	Spherical Mercator projections, graticule mathematics, boundary parsing	D3-Geo v3.1 / Topo v3
Satellite Basemaps	**Esri World Imagery API**	High-resolution global orbital satellite photography (Zero API Key)	ArcGIS REST Tile
Tactical Cartography	**Esri Dark Gray Canvas**	High-contrast military dark tactical vector basemap for night ops	ArcGIS REST Tile
Highway Vector Engine	**OpenStreetMap / Overpass**	Real road network coordinate extraction & verified shelter node geometries	OSM ODbL / Overpass
Real Telemetry Ingestion	**Open-Meteo REST API**	Live 60-second polling of 18 vulnerable stations across 6 Indian zones	REST JSON / HTTPS
Ensemble Predictions	**Open-Meteo GFS Ensemble**	30-member GFS ensemble processing for Extreme Forecast Index (EFI)	30-Member GFS
Telemetry Validation	**NASA POWER (MERRA-2)**	Independent hourly satellite reanalysis cross-checking live station data	MERRA-2 Product
Satellite Replay Sync	**NASA GIBS EOSDIS**	True Color MODIS Terra/Aqua satellite imagery synced to historic tracks	WMTS Archive
Disaster Ground Truth	**NOAA IBTrACS v04r00**	International Best Track ground truth for cyclonic tracks and pressures	NCEI Archive v04
Demographic Exposure	**Census of India 2011**	Sub-district population density figures for 5km exposure calculations	Open Govt Data India
Auditing & Security	**SHA-256 Crypto Hashing**	Immutable chronological audit trail of operator logins and dispatches	Browser WebCrypto
Offline Generation	**Python 3.14 / FPDF**	Automated system dossier & technical approach PDF compilation pipeline	Python 3.14 / FPDF
Cloud Deployment	**Vercel Edge Platform**	Global edge delivery, atomic deployments, SPA rewrite routing	Vercel Production

SECTION 3: END-TO-END FIVE-TIER SYSTEM ARCHITECTURE

Project VARUNA integrates five tightly coupled subsystems into an autonomous operational pipeline:

SYSTEM ARCHITECTURE SPECIFICATION

1. INGESTION & DATA FUSION TIER:
   - Telemetry Poller: Ingests 18 parameters every 60s from 18 stations (temperature, dew point, MSLP, wind gust, precipitation).
   - GFS 30-Member Ensemble Ingestion: Computes forecast spread and Extreme Forecast Index (EFI) up to 120 hours ahead.

2. ANOMALY DETECTION & CROSS-VERIFICATION TIER:
   - Dual-Source Agreement: Computes delta between Open-Meteo and NASA POWER MERRA-2. If confidence > 85%, sensor telemetry is verified.
   - Automated Z-Score Trigger: Compares readings against 30-year day-specific climatological normal. Auto-promotes station to WATCH (Z >= 2.5) or WARNING (Z >= 3.5).

3. 4D SPATIO-TEMPORAL RECONSTRUCTION TIER:
   - Trajectory Interpolation: 5 to 8 temporal steps per disaster benchmark calculating central pressure deficit and gale radius.
   - NASA GIBS Orbital Sync: Dynamically links MODIS Terra/Aqua True Color imagery corresponding to exact historical dates.

4. HYPER-LOCAL PINPOINT & BULLETIN TIER:
   - 5km Pinpoint Impact Polygon: Calculates exact spatial footprint (78.54 sq km) versus conventional 25km radius (1,963.5 sq km).
   - Automated Bulletin Synthesizer: Generates bilingual IMD-compliant 4-tier bulletins and SMS/voice broadcasts in 5 Indian languages.

5. EVACUATION & LOGISTICS COMMAND TIER:
   - Real Highway Network Overlay: Real National Expressways (Green) and State Highways (Amber) rendered over Esri Satellite/Dark imagery.
   - Capacity Matching: Compares Census population against real OSM shelter capacities. Flags secondary overflow if shortfall occurs.
   - Live Convoy Dispatch: Simulates relief supply trucks (Food, Water, Medical, Tarpaulins) with live speeds, routes, and ETAs.

SECTION 4: MATHEMATICAL FORMULATIONS & CORE ALGORITHMS

4.1 Hyper-Local Pinpoint Radius & False-Alarm Area Reduction (FAR)

Conventional district-level warnings define an alert footprint radius R_conv = 25 km, yielding an area A_conv = pi * (25)^2 = 1,963.5 sq km. For broader 50 km advisories, A_conv = 7,853.9 sq km. VARUNA isolates the immediate high-risk impact zone to R_varuna = 5 km, yielding A_varuna = pi * (5)^2 = 78.54 sq km.

Percentage False-Alarm Area Reduction (FAR_reduction) is quantitatively derived as:
FAR_reduction = [1 - (A_varuna / A_conv)] * 100
Against 25 km district warning: FAR_reduction = [1 - (78.54 / 1963.5)] * 100 = 96.0% reduction.
Against 15 km localized warning: FAR_reduction = [1 - (78.54 / 706.8)] * 100 = 88.9% reduction.

4.2 Dual-Source Atmospheric Confidence Index (ACI)

To eliminate false alarms caused by sensor degradation, VARUNA cross-checks dual live feeds (S1 = Open-Meteo, S2 = NASA POWER):
delta_P = | P_S1 - P_S2 | / P_S1 | delta_V = | V_S1 - V_S2 | / max(V_S1, 10) | delta_T = | T_S1 - T_S2 | / max(T_S1, 10)
ACI = 1.0 - [0.40 * min(delta_P, 1.0) + 0.40 * min(delta_V, 1.0) + 0.20 * min(delta_T, 1.0)]
Telemetric data is approved for bulletin generation only when ACI >= 0.85.

4.3 Anomaly Z-Score Formulation

Real-time observations x_t are standardized against 30-year day-of-year normal baselines (mu_d, sigma_d):
Z_t = (x_t - mu_d) / sigma_d
Automated System States: NORMAL (Z < 1.50) -> ELEVATED (1.50 <= Z < 2.50) -> ACTIVE WATCH (2.50 <= Z < 3.50) -> SEVERE WARNING (Z >= 3.50).

4.4 Evacuation Allocation & Capacity Shortfall Matching Algorithm

Evacuation population demand P_evac is computed from Census 2011 district density rho_d:
P_evac = A_varuna * rho_d * f_vulnerability (where f_vulnerability = 0.65 in coastal zones).
Available shelters {S_1, ... S_k} within search radius have rated capacities {C_1, ... C_k}.
Population is assigned greedily by road network distance d(origin, S_j).
Shortfall condition: If sum(C_j) < P_evac, Shortfall = P_evac - sum(C_j). The system automatically activates secondary overflow corridors.

SECTION 5: EVACUATION & RELIEF OPERATIONS COMMAND SPECIFICATION

The Evacuation & Relief Ops module (/evacuation) delivers actionable logistics capabilities on real cartography, replacing abstract warnings with tangible evacuation corridors and fleet tracking:

EVACUATION & RELIEF OPS ARCHITECTURE

1. REAL CARTOGRAPHIC ENGINE (ZERO API KEYS REQUIRED):
   - Esri World Satellite Imagery: High-resolution orbital photography loaded via ArcGIS REST tile servers.
   - Esri Dark Gray Tactical Canvas: Military dark vector basemap designed for high-contrast command rooms.
   - Real OpenStreetMap Highway Vectors: National and State Highway polylines with accurate geographic alignment.

2. FIVE REAL UPCOMING FORECAST THREATS (OCTOBER 5-8, 2026):
   - Threat 1: Bay of Bengal Cyclonic Depression Watch (85% Probability) | Dhamra-Digha-Sagar Corridor | 54,200 evacuees.
   - Threat 2: Western Ghats / Konkan Extreme Deluge (78% Probability) | Mahad-Chiplun Savitri River Basin | 41,000 evacuees.
   - Threat 3: North Gujarat / Kutch Storm Surge Watch (74% Probability) | Mandvi-Mundra Coastal Belt | 34,500 evacuees.
   - Threat 4: Coromandel Coast Deep Depression (81% Probability) | Cuddalore-Nagapattinam Cauvery Delta | 46,000 evacuees.
   - Threat 5: Brahmaputra Basin Inundation Watch (76% Probability) | Dibrugarh-Majuli Riverine Corridor | 31,000 evacuees.

3. FOUR-TIER COLOR-CODED EVACUATION HIGHWAY NETWORK:
   - Neon Emerald Green (#10b981): Primary High-Capacity Expressways (NH-116B, NH-66, NH-41, ECR, NH-15) with moving chevrons.
   - Neon Amber (#f59e0b): Secondary State Arterial Highways (SH-4, SH-96, SH-48, NH-36, AT Road) for civic bus fleets.
   - Vivid Cyan (#06b6d4): Coastal evacuation feeder links connecting beachside settlements.
   - Flashing Crimson Red (#ef4444): Lowland causeways marked 'CLOSED - INUNDATION RISK / DO NOT ENTER'.
   - Deep Purple (#818cf8): Secondary overflow highway routing to inland mega-shelters (e.g., Contai Mega Haven, Cap: 18,000).

4. ANIMATED EVACUATION FLOW PARTICLES & FLEET CONVOYS:
   - Continuous glowing particle streams flowing from Sector Alpha, Bravo, and Charlie directly into designated shelters.
   - Convoy Tracker simulating relief trucks with live cargo weights, speed (km/h), and countdown ETAs.

SECTION 6: COMPLETE WEBSITE NAVIGATION DIRECTORY & LIVE ROUTES

Screen 1: Tactical Access Gateway (https://varuna-access.vercel.app/#login)
2FA authentication, role permissions, live IST clock, one-click demo credentials.

Screen 2: Synoptic Telemetry & Anomaly Tracker (https://varuna-access.vercel.app/#dashboard)
18 live stations across 6 Indian zones, Open-Meteo vs NASA POWER cross-verification.

Screen 3: 4D Spatio-Temporal Track Replay (https://varuna-access.vercel.app/#track)
7 historic flagship disaster benchmarks with real NASA MODIS satellite synchronization.

Screen 4: Hyper-Local Alert Zone & IMD Bulletin (https://varuna-access.vercel.app/#bulletin)
5km pinpoint polygon, 75-85% false alarm reduction, official bilingual bulletins, 5-language SMS.

Screen 5: AI Predictor (Ensemble Outlook) (https://varuna-access.vercel.app/#outlook)
30-member GFS ensemble processing for Extreme Forecast Index (EFI) 1-5 days ahead.

Screen 6: Evacuation & Relief Operations Command (https://varuna-access.vercel.app/#evacuation)
Esri Satellite/Dark mapping, 4-tier colored highways, 5 active Oct 2026 threats, convoy fleet.

Screen 7: Data Sources & Methodology Dossier (https://varuna-access.vercel.app/#methodology)
Full disclosure of all 9 APIs, mathematical formulations, and PDF export.

Screen 8: Operational Security & System Settings (https://varuna-access.vercel.app/#settings)
API quota meters, network latency gauges, cryptographic audit logs, and RBAC policies.

Global Utility: Ask VARUNA (Accessible across all screens)
Grounded AI assistant answering operator questions using live telemetry and active alerts.

SECTION 7: COMPETITIVE ADVANTAGE, SECURITY POSTURE & SCALABILITY

WHY VARUNA WINS SIH 2026

1. Complete Alignment with Problem Statement SIH26078: Provides the exact multi-hazard early warning, risk communication, and evacuation capabilities demanded by MoES and NCMRWF.
2. Zero API Key Vulnerability: All mapping, telemetry, reanalysis, and satellite feeds operate on public, proxy-authenticated, or secure backend endpoints. Zero private keys are exposed to client browsers.
3. Production Cloud Deployment: Fully deployed on Vercel Global Edge Network with sub-second asset delivery worldwide.
4. Factually Grounded & Verifiable: Built on genuine meteorological physics, Census 2011 demographics, and NOAA/NASA archives.